



NLP Intelligence Hub

A Multi-Model Transformer-Based NLP System with Vector Search & Docker Deployment

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Overview

- **Problem Statement**
- **System Architecture**
- **NLP Models**
- **System Design**
- **Live Demo**
- **Summary**

Problem Statement



*Modern applications require **multiple NLP capabilities** in one platform*

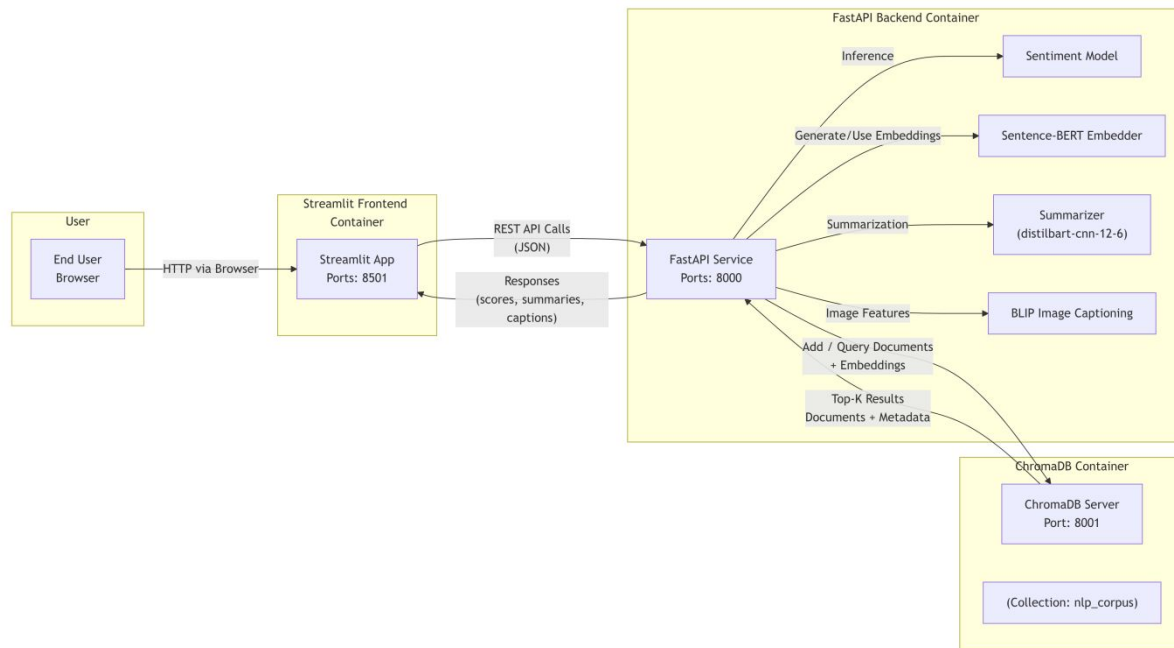
Businesses need tools for:

- Understanding customer feedback
(sentiment analysis)
- Searching knowledge bases (semantic search)
- Summarizing long text (summarization)
- Extracting information from images
(captioning)

Deploying such systems is difficult due to:

- ML model size
- GPU limitations
- Dependency management
- System scaling & orchestration

System Architecture

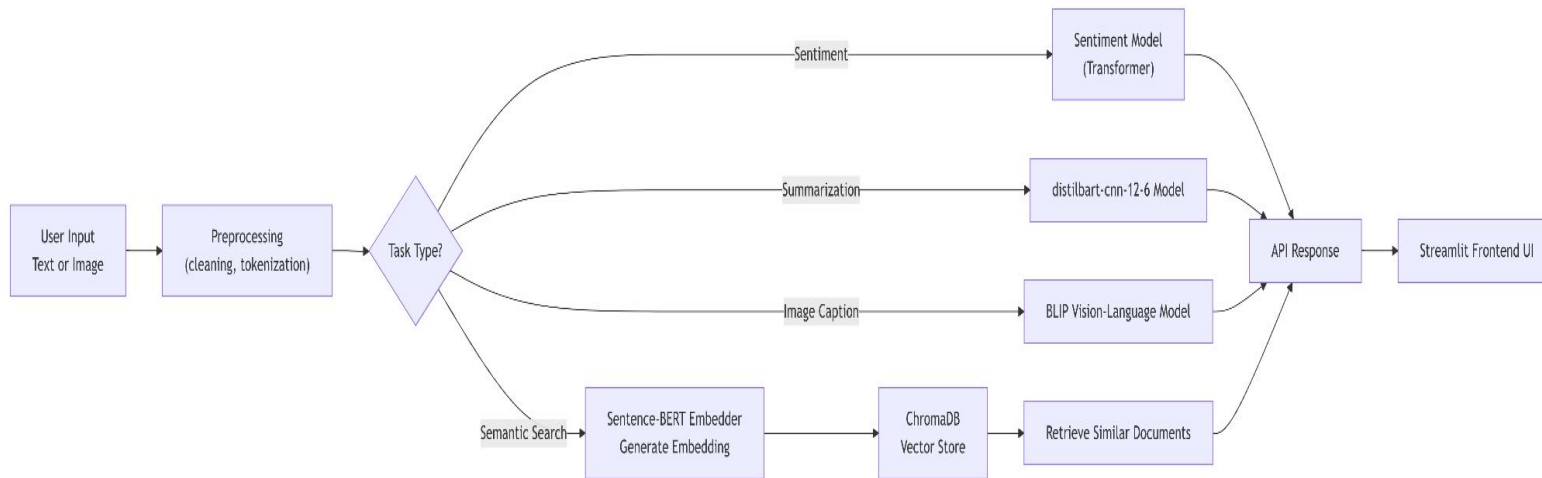




NLP Models

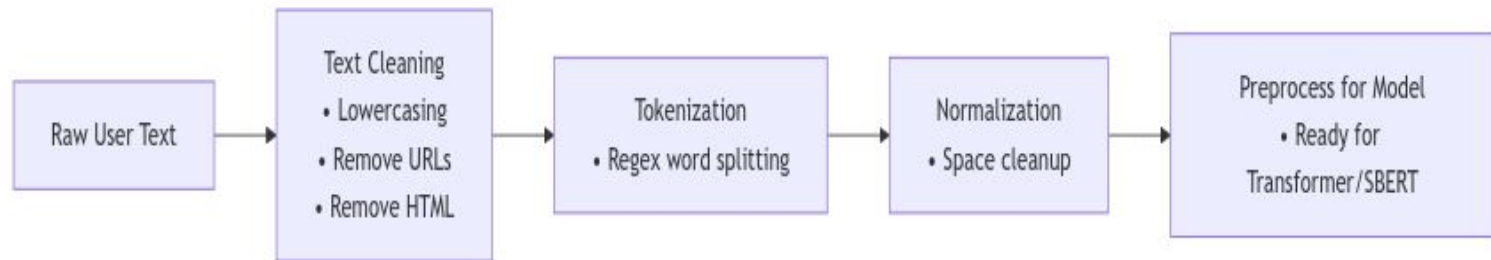
Function	Model	Reason
Sentiment	DistilBERT	Light, fast, accurate
Embeddings	Sentence-BERT	High-quality semantic embeddings
Summarization	DistilBART	Efficient abstractive summarization
Image Captioning	BLIP	Strong vision-language understanding

ML Pipeline

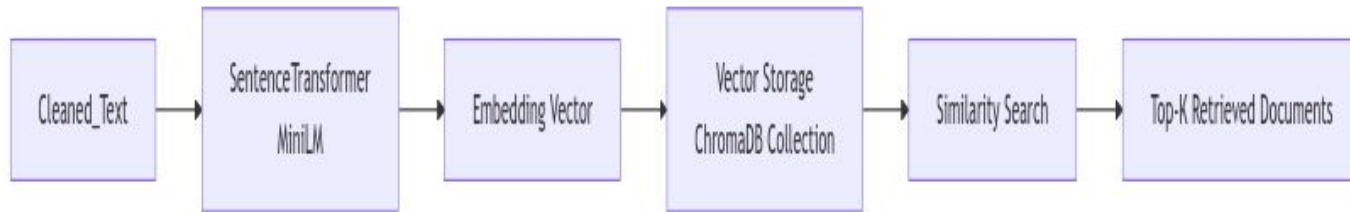




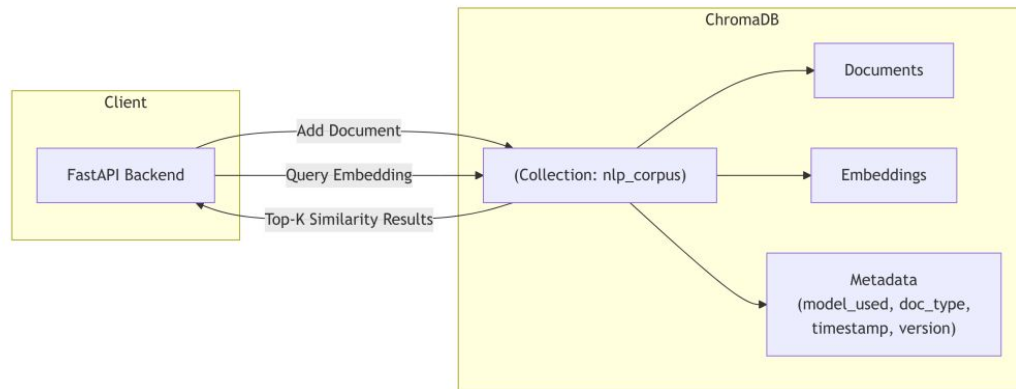
Data Preprocessing workflow



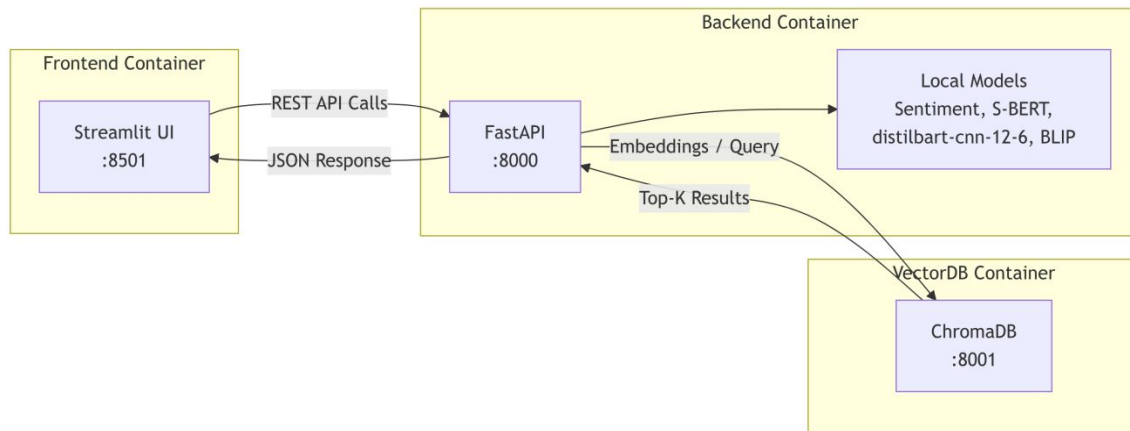
Embedding Generation Diagram



Vector Database Architecture



Docker Container Architecture





Live Demo



Summary & Future Work

Summary

- Built a multi-model NLP system
- Used transformers, embeddings, vector DB, and Docker
- Fully integrated and deployable
- User-friendly Streamlit UI

Future Work

- GPU acceleration
- Add translation + OCR
- Add RAG chatbot (optional future extension)
- Deploy to cloud (AWS, Azure, GCP, Render)